Danny **Fryer** PhD

SENIOR DATA SCIENTIST | DATA INFRASTRUCTURE & APPLIED ML

Melbourne, Australia | danielvfryer@gmail.com

Website | LinkedIn | GitHub: frycast | Google Scholar

PhD data scientist building reliable data systems and deploying statistical and ML models in practice. Combines hands-on engineering with technical leadership, cross-functional collaboration and award-winning teaching.

EDUCATION

PhD, Statistics
The University of Queensland
Jan 2019 - Apr 2023
Understanding and explaining machine learning models.

BSc, First Class Honours
Mathematics and Statistics
La Trobe University
2014 - Nov 2018

EXPERTISE

Data systems
Data integration and infrastructure
SQL and relational modelling
Operational workflow design
Power BI and Shiny dashboards

Statistics & machine learning
Statistical and predictive modelling
Explainable AI and Shapley values
Survey design and analysis
R and research software

Leadership & enablement
Technical roadmap development
Cross-functional collaboration
Agentic AI development
AI adoption, training and policy
Teaching and technical workshops

RECOGNITION

David Myer's Medal | 2018
La Trobe University
Top honours student in Science, Health and Engineering.

Vice-Chancellor's Sessional Teaching Award | 2020
La Trobe University

Future Superstar Award | 2021
Faculty of Science
The University of Queensland

Associate Fellow | 2021
Higher Education Academy (AFHEA)

EXPERIENCE

Senior Data Scientist (Data Infrastructure)
The Social Research Centre
Jun 2023 - Present

- Led delivery of a multi-system data integration and analytics platform that doubled throughput and remained stable through three major infrastructure migrations.
- Owned business-critical incentives, unsubscribe and recovery workflows; built interfaces to close operational gaps.
- Deployed a production propensity model to predict participant completion; developed SQL data models, Power BI reporting and Shiny dashboards.
- Guided platform priorities with engineering, operations and executives; advised on AI adoption, training and policy.

Data Scientist | The Social Research Centre
Aug 2021 - Jun 2023

Associate Lecturer / Adjunct Lecturer
La Trobe University
Associate: Jan - Dec 2023 | Honorary adjunct: Dec 2023 - Present

- Designed and coordinated statistics and data science subjects, including probability and meta-analysis; developed accessible, rigorous teaching and assessments.

Earlier: Sessional Lecturer, Jul 2018 - Jan 2023; database instruction, Mar 2018 - Jan 2022.

Course Instructor & Author
The University of Auckland | NZ Social Statistics Network
Nov 2018 - Present

- Authored a four-day SQL and Integrated Data Infrastructure course and textbook; delivered the course over 14 times to government, research and academic audiences.

Research Statistician
The University of Queensland / Monash University
UQ: Jan 2019 - Jan 2022 | Monash: Oct 2020 - Jan 2022

- Developed and tested acute kidney injury diagnostic algorithms using global ISARIC clinical data; consulted on survey design and analysis in politics and international relations.

RESEARCH & OPEN SOURCE

Research on Shapley values, feature selection and uncertainty in model explanations, published in PeerJ Computer Science and Frontiers in Applied Mathematics and Statistics.

rcosmo - Lead developer of the CRAN R package for cosmological data analysis (La Trobe, Jun 2018 - Feb 2019); published in The R Journal, with a 2020 Australian Astrostatistics Association commendation.